

Memory Management

CS232 Operating Systems

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Early Systems

- OS was just a (bunch of) libraries
- Only one user program ran

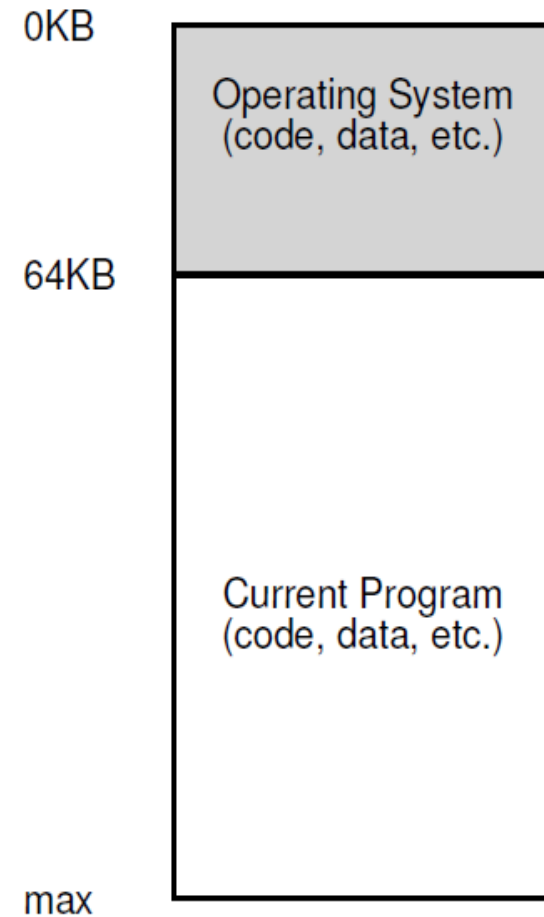


Figure 13.1: **Operating Systems: The Early Days**

Multiprogramming and Time Sharing

- Machines were expensive.
- Efficient use demanded sharing
 - b/w processes
 - b/w users
- With many users interactivity became important
- Can we give all the memory to one process?

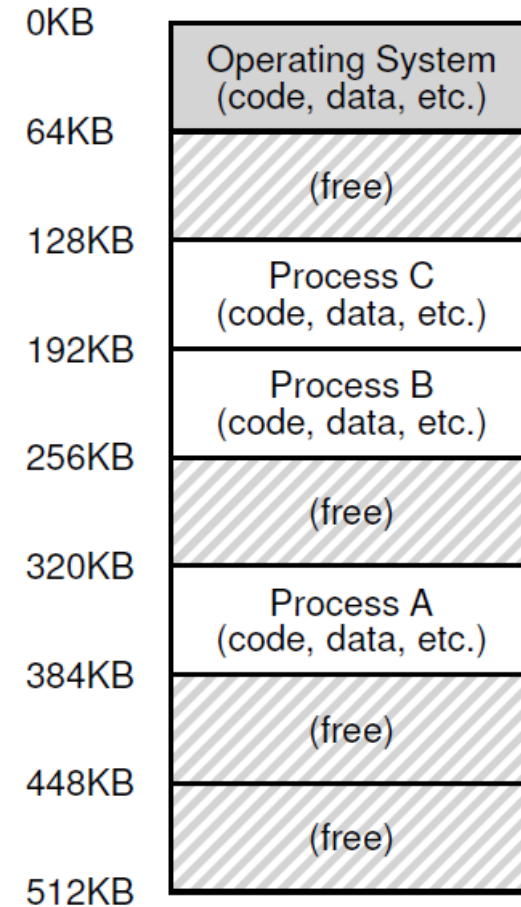


Figure 13.2: **Three Processes: Sharing Memory**

Address Space

- It's an abstraction
 - a running program's view of memory
- The program thinks it's loaded at address 0
- But to take example of process A, it's loaded at address 320KB

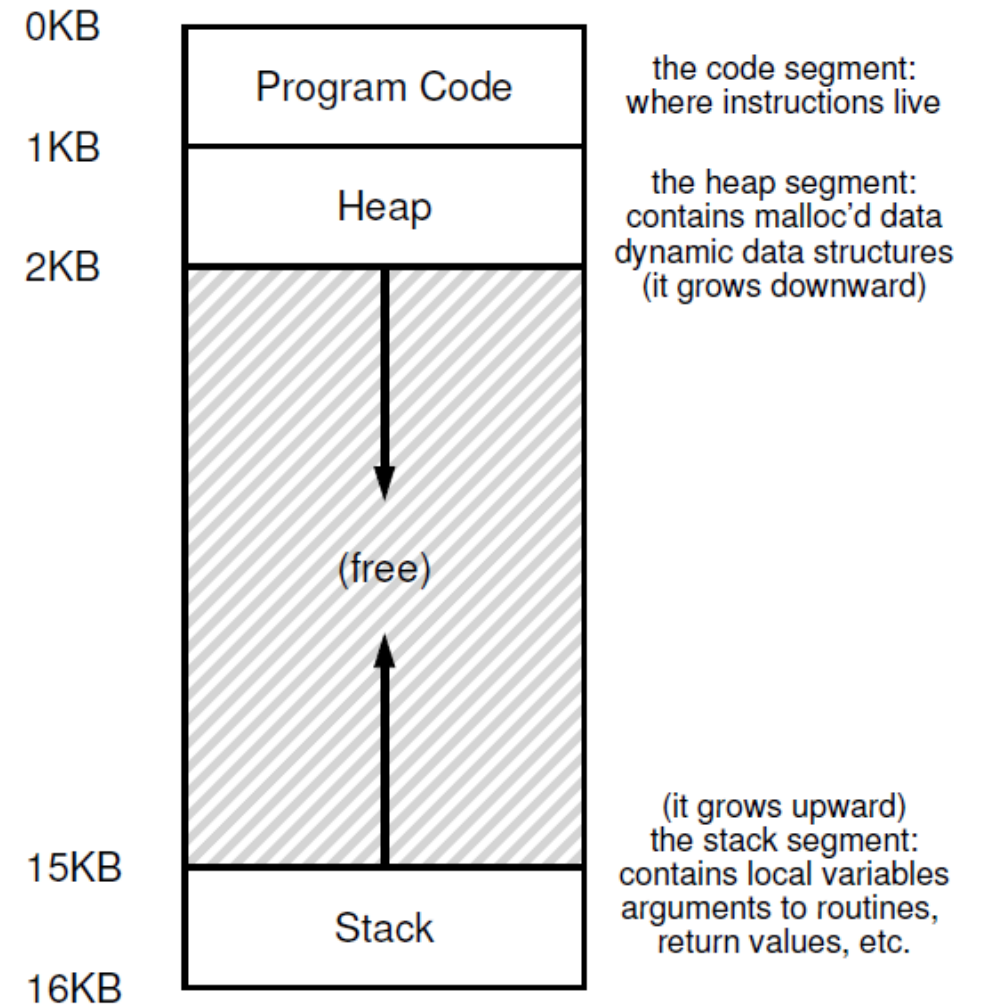


Figure 13.3: An Example Address Space

Goals

- Transparency
- Efficiency
 - Time
 - Space
- Protection
 - Isolation

Pointers store virtual addresses

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  int main(int argc, char *argv[]) {
4      printf("location of code   : %p\n", (void *) main);
5      printf("location of heap   : %p\n", (void *) malloc(1));
6      int x = 3;
7      printf("location of stack : %p\n", (void *) &x);
8      return x;
9  }
```

When run on a 64-bit Mac, we get the following output:

```
location of code   : 0x1095afe50
location of heap   : 0x1096008c0
location of stack  : 0x7fff691aea64
```